

## Solution Code - Spring Core and Maven - Annotation\_AOP\_Logging

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Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

### Source File: src\main\java\com\library\AppConfig.java

```
1 | package com.library;
2 |
3 | import org.springframework.context.annotation.*;
4 |
5 | @Configuration
6 | @EnableAspectJAutoProxy
7 | @ComponentScan(basePackages = "com.library")
8 | public class AppConfig {}
```

### Source File: src\main\java\com\library\BookService.java

```
1 | package com.library;
2 |
3 | import org.springframework.stereotype.Service;
4 |
5 | @Service
6 | public class BookService {
7 |     public void performCheckout() {
8 |         System.out.println("Processing user book checkout transaction...");
9 |     }
10 | }
```

### Source File: src\main\java\com\library\LibraryApp.java

```
1 | package com.library;
2 |
3 | import org.springframework.context.ApplicationContext;
4 | import org.springframework.context.annotation.AnnotationConfigApplicationContext;
5 |
6 | public class LibraryApp {
7 |     public static void main(String[] args) {
8 |         ApplicationContext context = new AnnotationConfigApplicationContext(AppConfig.class);
9 |         BookService service = context.getBean(BookService.class);
10 |         service.performCheckout();
11 |     }
12 | }
```

### Source File: src\main\java\com\library\LoggingAspect.java

```
1 | package com.library;
2 |
3 | import org.aspectj.lang.annotation.Aspect;
```

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```
4 | import org.aspectj.lang.annotation.Before;
5 | import org.springframework.stereotype.Component;
6 |
7 | @Aspect
8 | @Component
9 | public class LoggingAspect {
10 |     @Before("execution(* com.library.BookService.*(..))")
11 |     public void logBefore() {
12 |         System.out.println("[ANNOTATION-AOP-LOG]: Method invocation monitored and logged successfully.");
13 |     }
14 | }
```